

化學濾網與循環經濟

Bring Chemical Filters into Circular Economy

Jason Liu

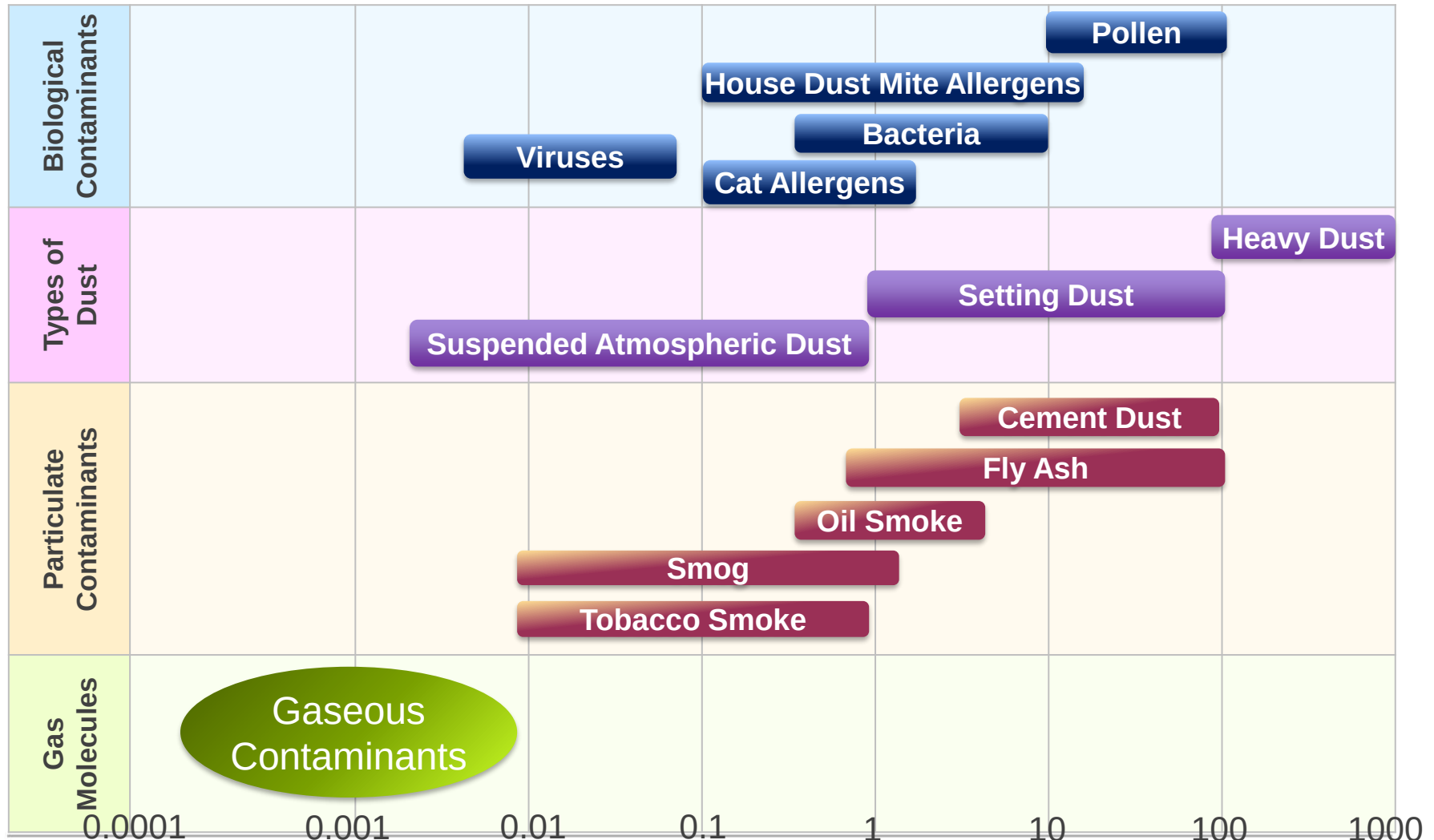
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CONNECT

What is AMC?

What is AMC?

Migration of Filtration Targets in Clean Environments

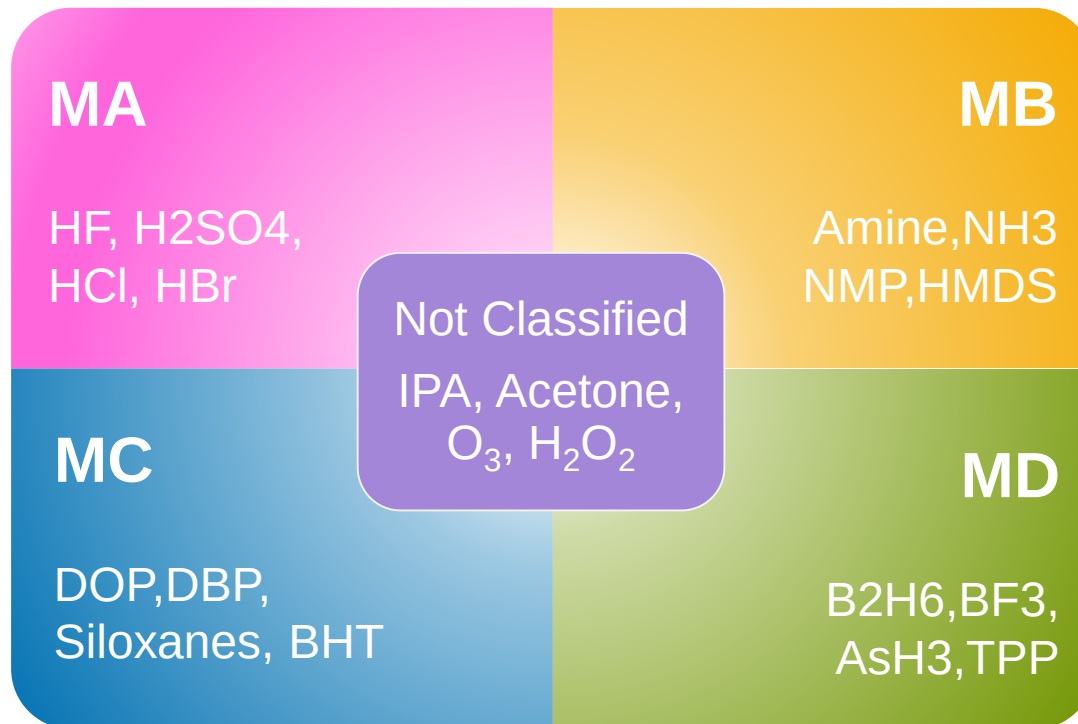


What is AMC?

- **A**irborne **M**olecular **C**ontamination
 - chemical contamination
 - in the form of vapors
 - not relevant to the cleanroom classification
 - effects on semiconductor process
 - haze on wafer or lens
 - corrosion, T-topping
 - ...

What is AMC?

Classification of AMC Levels



SEMI F21-1102

Molecular **A**cid
Molecular **C**ondensable

Molecular **B**ase
Molecular **D**opant

What is AMC?

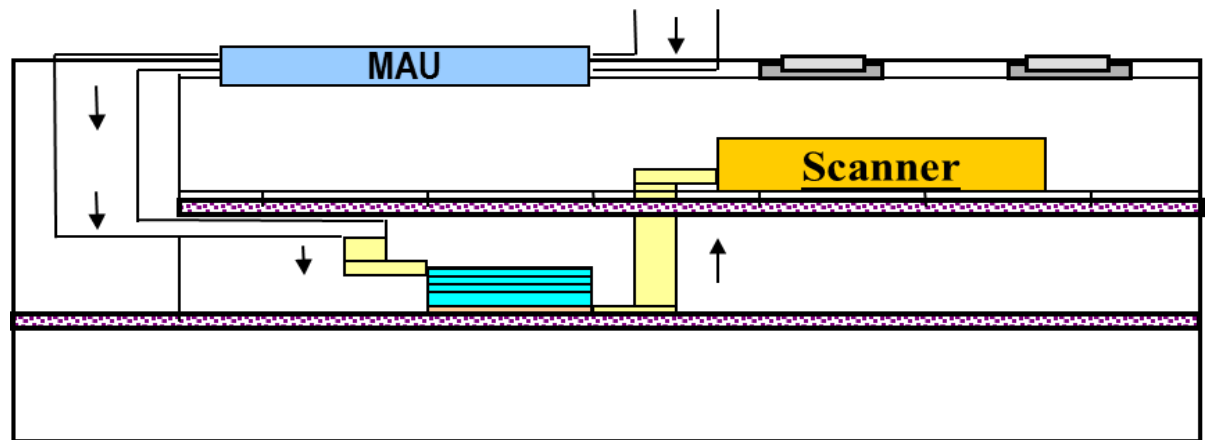
Sources of AMC

- Inside the Cleanroom
 - Interior / building materials
 - Instrument / parts materials
 - Process chemicals
 - Operating persons
- Outside the cleanroom
 - Auto & Motor exhaust gas
 - Factory exhaust gas
 - General environment

AMC Solutions

AMC Solutions

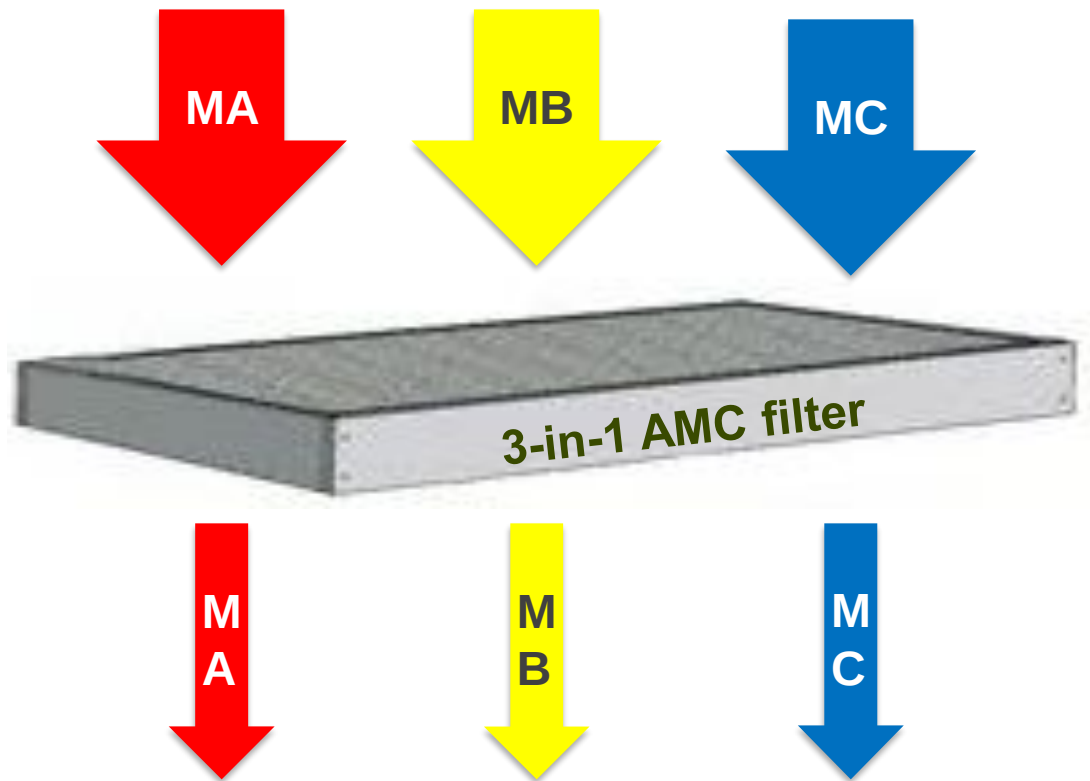
- Decrease upstream AMC
 - Air washer
- Remove downstream AMC
 - Complete Air Supply



AMC Solutions

Chemical (AMC) Filters

- How the AMC filter works



Env. AMC concentration:

MA: 10 ppb

MB: 10 ppb

MC: 10 ppb

AMC outlet:

MA: <1 ppb

MB: <1 ppb

MC: <1 ppb

AMC Solutions

Chemical (AMC) Filters

- Approx. Spec.

AMC Filter Spec.	FFU 2x4	FFU 4x4
Length	80 cm	100 cm
Width	50 cm	90 cm
Height	20~30 cm	15~30 cm
Weight	10~20 kgs	15~35 kgs

- Filter structure
 - Aluminum frame
 - Faceguards
 - Filter cloths
 - Adsorption Materials
 - Supporting Nets



AMC Solutions

AMC (Chemical) Filters

- Annual filter waste



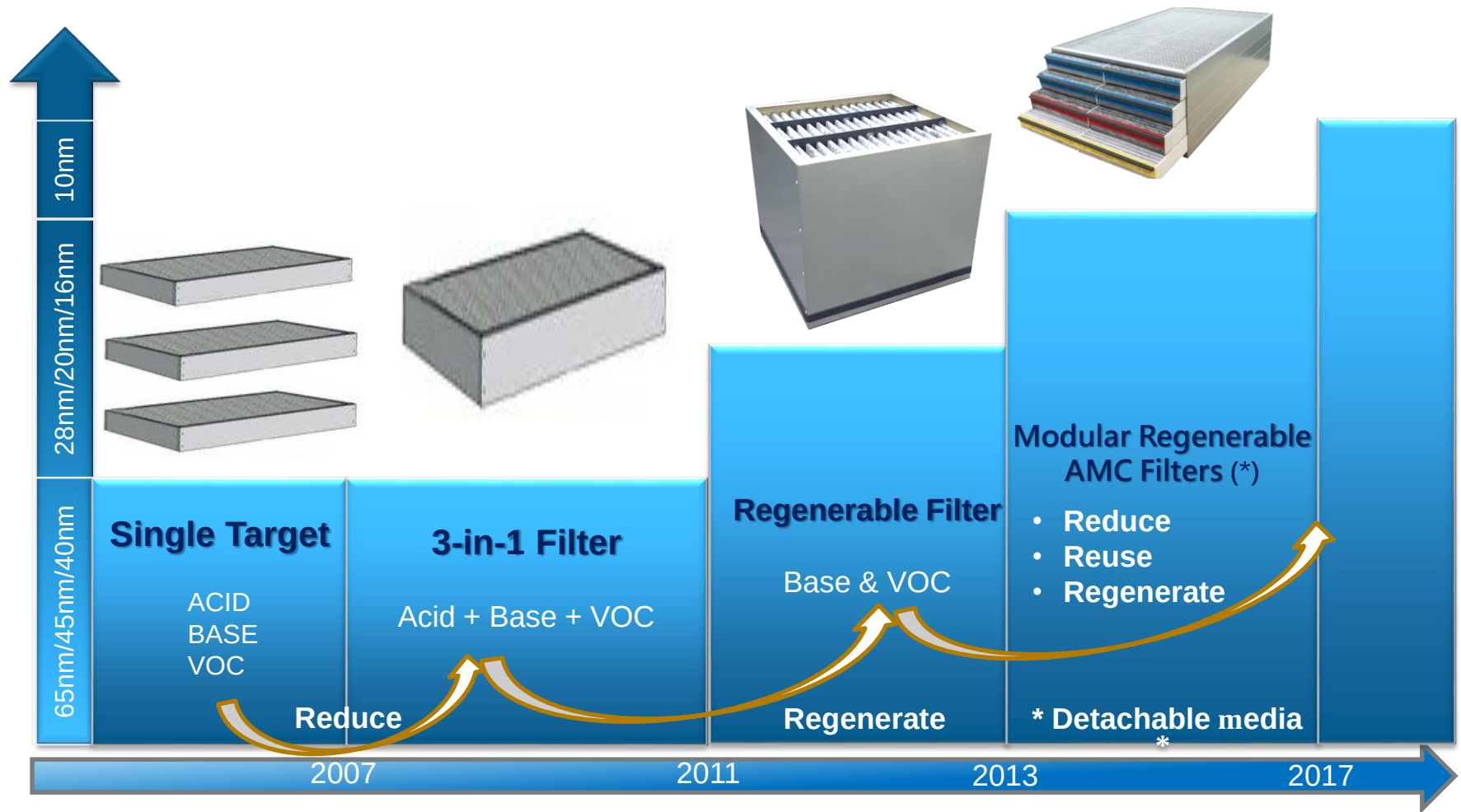
Filter type	Quantity (pc)	Height (m)
Box type	40,000	11,680
Panel type	50,000	10,000
Total	90,000	21,680

- Headaches on handling the filter disposals
 - Labor on exchanging filters
 - Cost of stocking the retired filters
 - Cost on dumping and processing filter disposals

Bring Chemical Filters into Circular Economy

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Roadmap of Tool/FFU Type AMC filter



Bring Chemical Filters into Circular Economy

A Breakthrough of AMC Filter Design

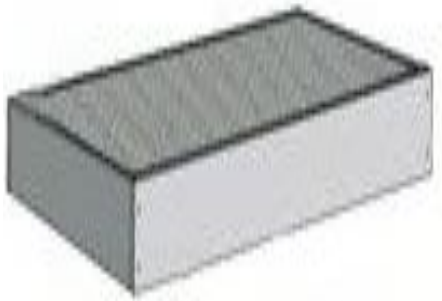


Modular panel type AMC filter
Designed & supplied by **GreenFILTEC**

Features :

- a) Replaceable media
- b) Regenerable media (*)
- c) Reusable frame
- d) Quick & easy installation
(save 30% installation time)
- e) Outgassing free (PU free)
- f) Low pressure drop (15 Pa/layer)
- g) Waste reduction

Bring Chemical Filters into Circular Economy



Traditional 3-in-1 AMC Filter

One time use

Non-replaceable,
Quality + Cost

Heavy

Long

Huge

Frame

Media

Weight

Installation Time

Waste

GreenFILTEC Modular AMC Filter

Reusable

Replaceable,
Quality + Cost + Env. Friendly

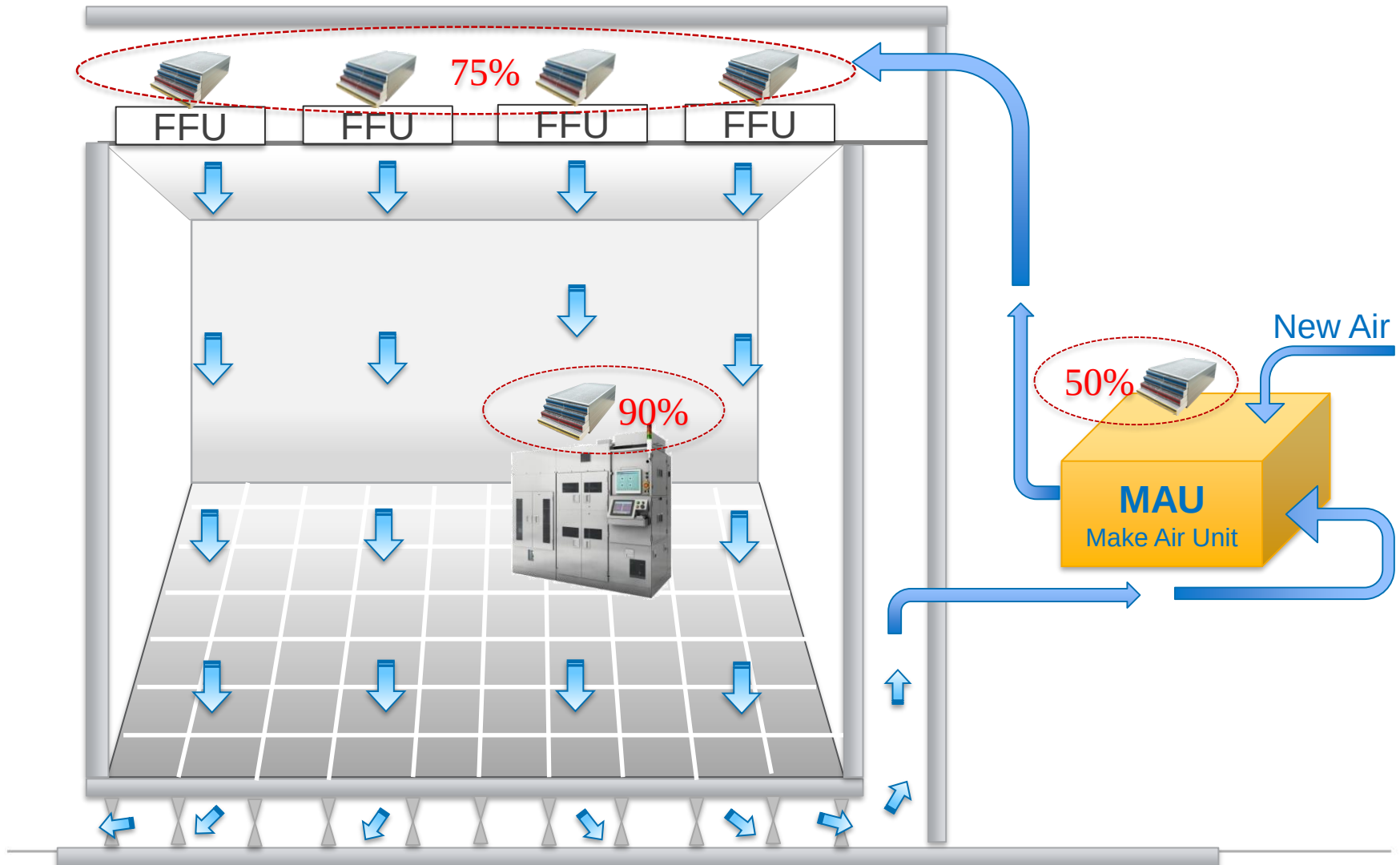
Light

Short (↓ 30%)

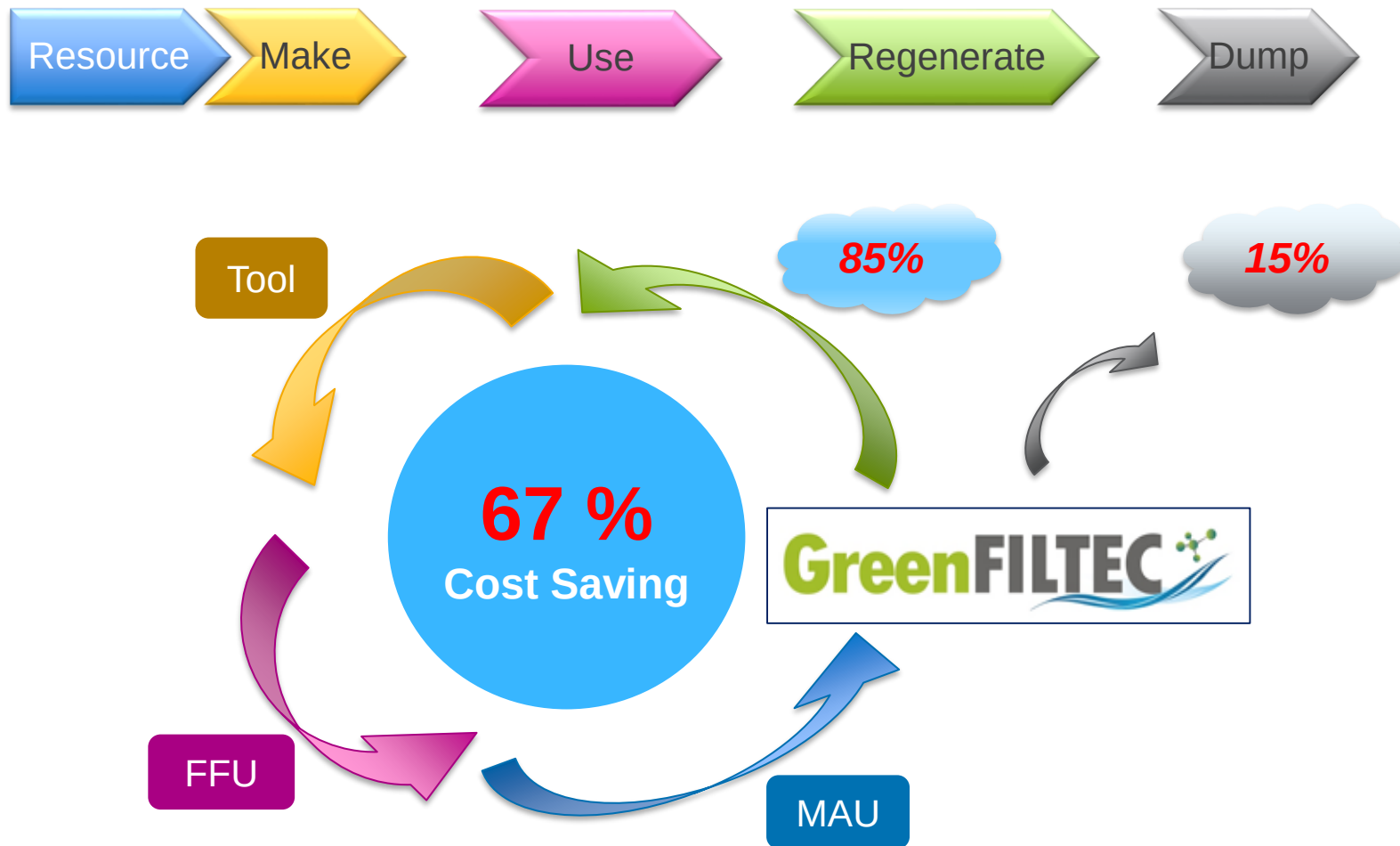
Little

Bring Chemical Filters into Circular Economy

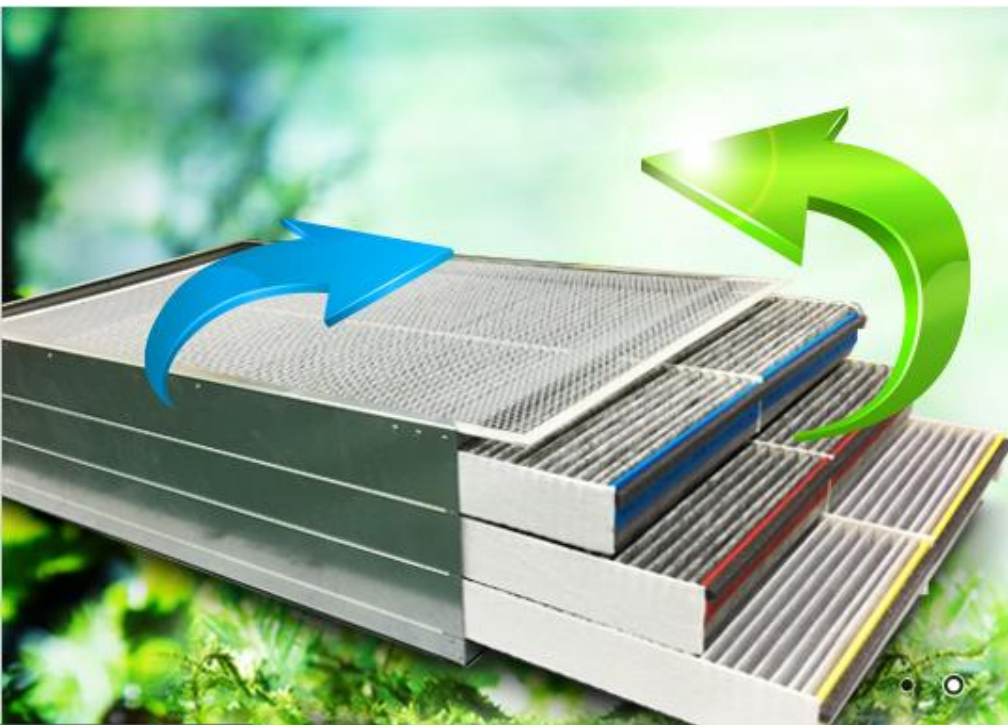
Maximize the Usage and Lifetime of AMC Filters



Bring Chemical Filters into Circular Economy



Go clean. Think Green.



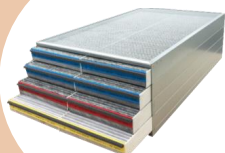
安全
Safety

節費
Cost-Saving

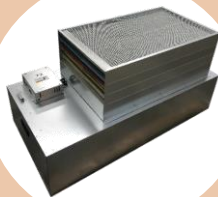
減廢
Waste
Reduction

Lead Air Filtration to GREEN Generation!
GreenFILTEC . Your best AMC consultant!

Services



**Modular
Chemical Filter**



ChemFFU



**Mobile Pollution
Filtration System**



**AMC
Sampling & Analysis**